二维数组练习2

人员

臧晨云 隋溟琛 霍佳铭 于昊华 刘子郴 何佩瑜 刘克阳 初星熠 陈之恒 刘奕男 李景翾 张高 到课

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3839

王向东老师周六一点半C++二维数组练习1

刷新

#	用户名	姓名	编程分	时间	A	B	C	D	E	F	G	H
1	chenzhiheng	陈之恒	800	21	100	100	100	100	100	100	100	100
2	zangchenyun	臧晨云	800	26	100	100	100	100	100	100	100	100
3	suihaochen	隋溟琛	800	28	100	100	100	100	100	100	100	100
4	huojiaming	霍佳铭	800	34	100	100	100	100	100	100	100	100
5	zhanggao	张高	730	23	100	100	100	100	100	100	100	30
6	hepeiyu	何佩瑜	700	19	100	100	100	100	100	100	100	
7	yuhaohua	于昊华	700	20	100	100	100	100	100	100	100	0
8	liuyinan	刘奕男	700	26	100	100	100	100	100	100	100	
9	lijingxuan	李景翾	700	28	100	100	100	100	100	100	100	
10	chuxingyi	初星熠	600	16	100	100	100	100	100	100		
11	liuzichen123	刘子郴	600	16	100	100	100	100	100	100	0	
12	liukeyang	刘克阳	600	24	100	100	100	100	100	100		

作业

https://cppoj.kids123code.com/contest/4029 (课上讲了 A ~ E 题, 作业是 A ~ E 题必做)

课堂表现

今天的第 5 题比较复杂一些, 同学们课上基本都没有做完这道题, 课下要再好好复盘一下这道题目。

课堂内容

两个数相邻吗?

```
#include<bits/stdc++.h>
using namespace std;
int n,m;
int a[105][105];
int c,b;
int bx,by,cx,cy;
int main()
{
    cin>>n>>m;
    for(int i=1;i<=n;i++)
```

```
    {
        for(int j=1;j<=m;j++)
        {
            cin>>a[i][j];
        }
    }
    cin>>b>>c;
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
        {
            if(a[i][j]==b)
            {
                bx=i;
                by=j;
            }
            if(a[i][j]==c)
            {
                cx=i;
                cy=j;
            }

        }
    }
    if((bx==cx&&abs(by-cy)==1)||(by==cy&&abs(cx-bx)==1))
    {
        cout<<"Y";
    }
    else cout<<"N";
}
```

地雷数量求解

```
#include <bits/stdc++.h>
using namespace std;
int a[205][205];
int cnt;
int main()
{
    int n,m;
    cin>>n>>m;
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
        {
            cin>>a[i][j];
            int x=0;
            while(a[i][j]>0)
            {
                x+=a[i][j]%10;
                a[i][j]/=10;
```

```
            }
            if(x%2!=0)
            {
                cnt++;
            }
        }
    }
    cout<<cnt;
}
```

计算鞍点

```
#include <bits/stdc++.h>

using namespace std;

int a[15][15];

int main()
{
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= 5; j++) {
            cin >> a[i][j];
        }
    }
    int cnt = 0, maxx, minn;
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= 5; j++) {
            maxx = 0;
            for (int k = 1; k <= 5; k++) {
                if (a[i][k] > maxx) {
                    maxx = a[i][k];
                }
            }
            minn = 9999999;
            for (int k = 1; k <= 5; k++) {
                if (a[k][j] < minn) {
                    minn = a[k][j];
                }
            }
            if (a[i][j]==maxx && a[i][j]==minn) {
                cout << i << " " << j << " " << a[i][j] << endl;
                cnt++;
            }
        }
    }
    if (cnt == 0) {
        cout << "not found" << endl;
    }
```

```
    return 0;
}
```

图像相似度

```
#include <bits/stdc++.h>
using namespace std;
int a[205][205],b[205][205];
double x;
int main()
{
    int n,m;
    cin>>n>>m;
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
        {
            cin>>a[i][j];
        }
    }
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
        {
            cin>>b[i][j];
        }
    }
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
        {
            if(a[i][j]==b[i][j])
            {
                x++;
            }
        }
    }
    cout<<fixed<<setprecision(2)<<1.0*x/(n*m)*100;
}
```

孤独的素数

```
#include <bits/stdc++.h>

using namespace std;

int a[55][55], b[55][55];
```

```
int main()
{
    int n, m;
    cin >> n >> m;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            cin >> a[i][j];
        }
    }

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            int cnt = 0;
            for (int k = 1; k <= a[i][j]; k++) {
                if (a[i][j]%k == 0) {
                    cnt++;
                }
            }
            if (cnt == 2) {
                b[i][j] = 1;
            }
            else {
                b[i][j] = 0;
            }
        }
    }

    int cntt = 0;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            if (b[i][j] == 1) {
                if (b[i-1][j-1]==0 && b[i-1][j]==0 && b[i-1][j+1]==0 && b[i][j-
1]==0 && b[i][j+1]==0 && b[i+1][j-1]==0 && b[i+1][j]==0 && b[i+1][j+1]==0) {
                    cntt++;
                }
            }
        }
    }
    cout << cntt << endl;
    return 0;
}
```

```
#include <bits/stdc++.h>

using namespace std;

int a[55][55], b[55][55];

int main()
{
    int n, m;
```

```
    cin >> n >> m;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            cin >> a[i][j];
        }
    }

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            int cnt = 0;
            for (int k = 1; k <= a[i][j]; k++) {
                if (a[i][j]%k == 0) {
                    cnt++;
                }
            }
            if (cnt == 2) {
                b[i][j] = 1;
            }
            else {
                b[i][j] = 0;
            }
        }
    }

    int cntt = 0;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            if (b[i][j] == 1) {
                int cnt = 0;
                for (int x = i-1; x <= i+1; x++) {
                    for (int y = j-1; y <= j+1; y++) {
                        if (b[x][y] == 1) {
                            cnt++;
                        }
                    }
                }
                if (cnt == 1) {
                    cntt++;
                }
            }
        }
    }
    cout << cntt << endl;
    return 0;
}
```